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HUJI Applicative Research Report — Exact & Social Sciences — Week 46, 2026

1 high-potential paper(s) · Exact & Social Sciences · Weekly · Week 46, 2026 · Generated November 09, 2026

This week's digest features a significant innovation in materials science from Professor David Avnir's research group. The team has developed 'prochiral nanocrystals,' demonstrating a novel ability to induce and precisely control chirality at the nanoscale. This breakthrough allows for the creation of advanced chiral materials with tailored properties. Such materials hold immense commercial potential across high-value sectors, from enabling more efficient and safer drug manufacturing to developing highly sensitive optical sensors and next-generation semiconductor components.

RESEARCHERS & RESEARCH SUBJECTS — HIGH-POTENTIAL PAPERS

- **David Avnir** — Materials, Drug Discovery, Other

EXACT & SOCIAL SCIENCES — RESEARCH HIGHLIGHTS

1. Prochiral nanocrystals. 28/50

"Prochiral nanocrystals offer tunable chirality, promising breakthroughs in drug synthesis, advanced sensors, and next-gen electronics."

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ABOUT THIS RESEARCH

The research focuses on the creation and manipulation of nanocrystals that lack inherent chirality but can be induced to exhibit chiral properties. This work likely explores how symmetry breaking at the nanoscale can be controlled for specific physical or chemical functions.

COMMERCIAL ANGLE

Controlling nanocrystal chirality offers significant potential for developing advanced chiral materials for enantiopure drug manufacturing, specialized optical sensors, and next-generation semiconductor components.

COMMERCIALISATION METRICS

Novelty 9/10

The development of prochiral nanocrystals represents a fundamental paradigm shift in materials science, enabling site-specific chemical reactivity and enantioselective catalysis on a nanometric scale. This platform-enabling technology creates a novel category of chiral building blocks with significant commercial potential in asymmetric synthesis and pharmaceutical manufacturing.

Commercial Potential 4/10

While prochiral nanocrystals have interesting applications in asymmetric catalysis and sensing, the lack of specific functional data or a defined problem statement makes it difficult to assess the scalability or market readiness of the technology. It remains a fundamental materials science advancement that requires significant downstream optimization to reach industrial or clinical utility.

Market Size 6/10

While prochirality is a foundational challenge in asymmetric catalysis and pharmaceutical manufacturing, the technology is currently at a fundamental material science stage with long-term, unquantified commercial pathing. The potential market value is high due to the pervasiveness of chirality in drug development, but the lack of demonstrated industrial scalability limits its current market-size score.

Tech Readiness 2/10

The study of prochiral nanocrystals is fundamental materials science focused on structural symmetry and surface chemistry, placing it firmly at the early benchtop proof-of-concept phase. There is no clear line of sight to a specific industrial or clinical application at this stage of research.

IP Strength 7/10

Prochiral nanocrystals represent a highly specialized materials science platform with significant potential for proprietary control over synthesis and surface modification, though actual defensibility is contingent upon the breadth of the specific composition-of-matter and application-specific patents.

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